

UNIT - IV

Dynamic Programming

Suboptimization and Principle of Optimality

The process of sub optimization was stated by Richard Bellman as the principle of optimality:

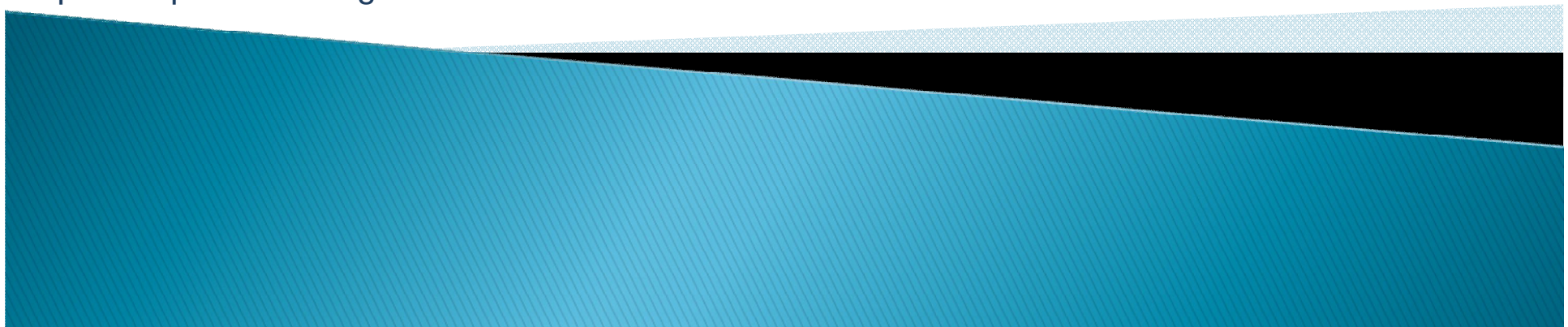
Bellman's Principle of Optimality

An optimal policy (or set of decisions) has the property that whatever the initial state and initial decision are the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

It implies that given the initial state of a system, an optimal policy for the subsequent stages does not depend on the policy adopted at the preceeding stages

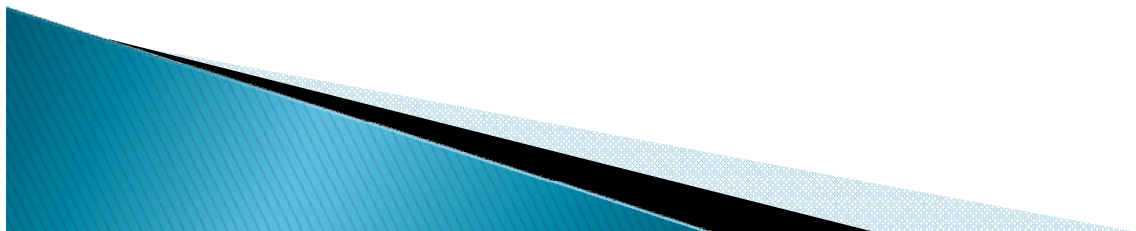
(OR)

Future decisions for the remaining stages will constitute an optimal policy regardless of the policy adopted in previous stages



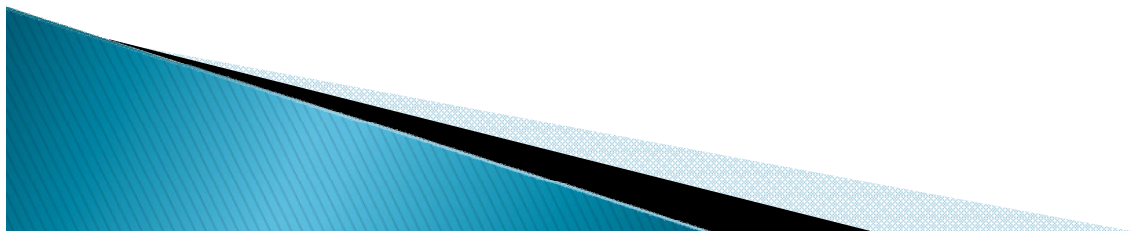
Principle of Optimality

- ▶ DP is an approach in which the problem is broken down into a number of smaller sub problems called *stages*.
- ▶ These sub problems are then solved sequentially until the original problem is finally solved.
- ▶ A particular sequence of alternatives (course of action) adopted by the decision-maker in a multi-stage decision problem is called a *policy*.



Dynamic Programming (DP)

- ▶ DP determines the optimum solution of a multivariable problem by decomposing it into n stages with each stage constituting a single-variable sub problem.
- ▶ The advantage of this decomposition is that
 - The optimization process at each stage involves one variable only.

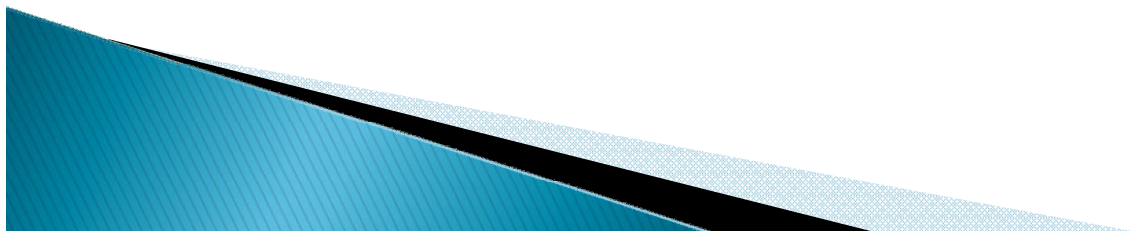


Shortest-Route Problem

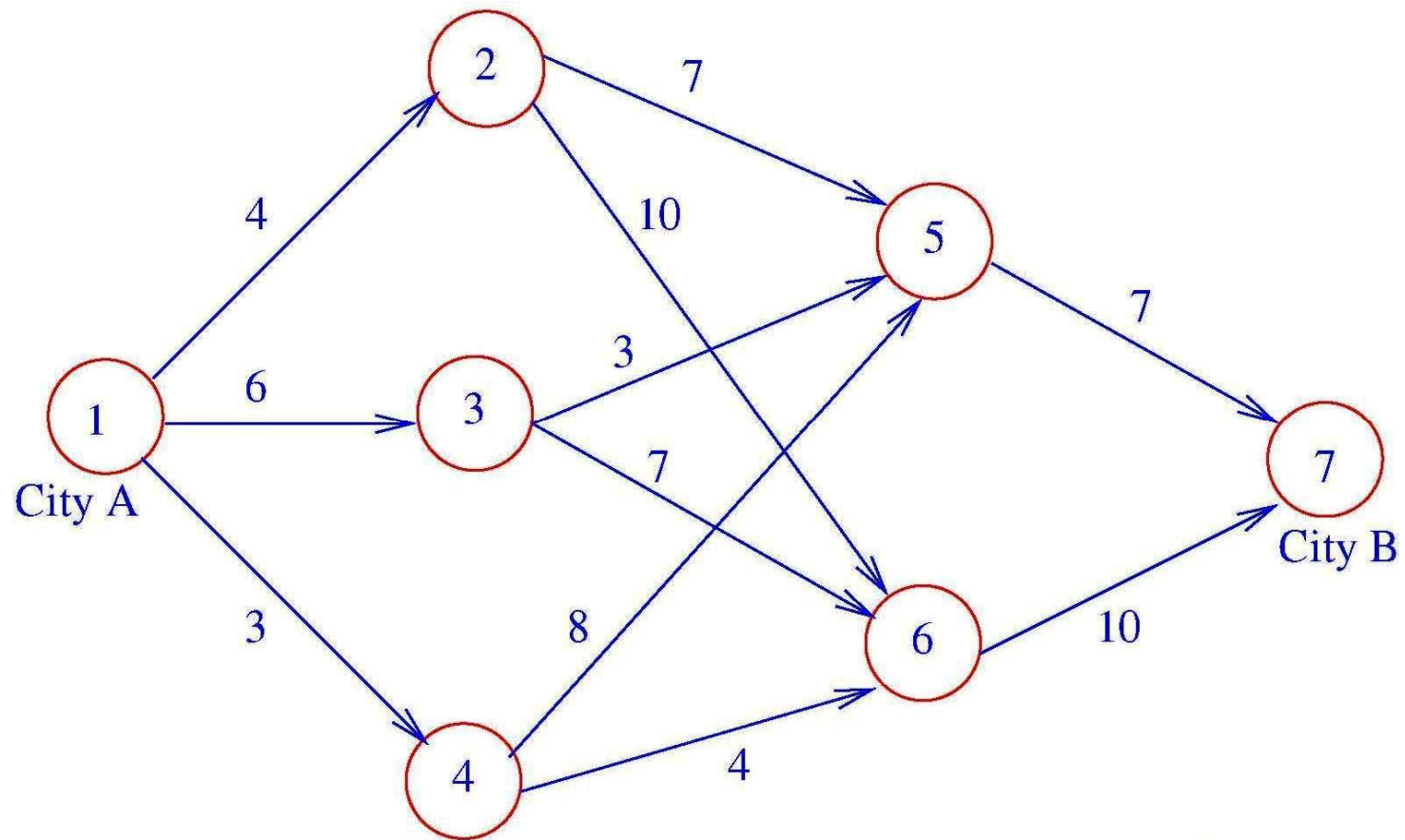
A salesman, located in a city A, decided to travel to city B. He knew the distances of alternative routes from city A to city B.

He then drew a highway network map as shown in the figure. The city of origin A is city 1. The destination city B is city 7. Other cities through which the salesman will have to pass through are numbered 2 to 6.

The arrow representing routes between cities and distances in kilometers are indicated on each route.

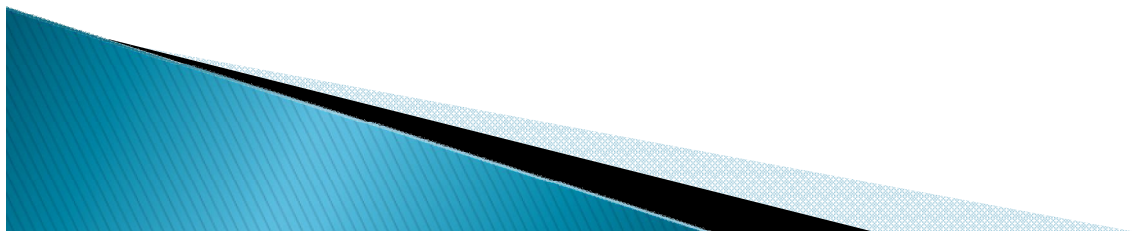


The salesman problem is to find the shortest route from city A to B.

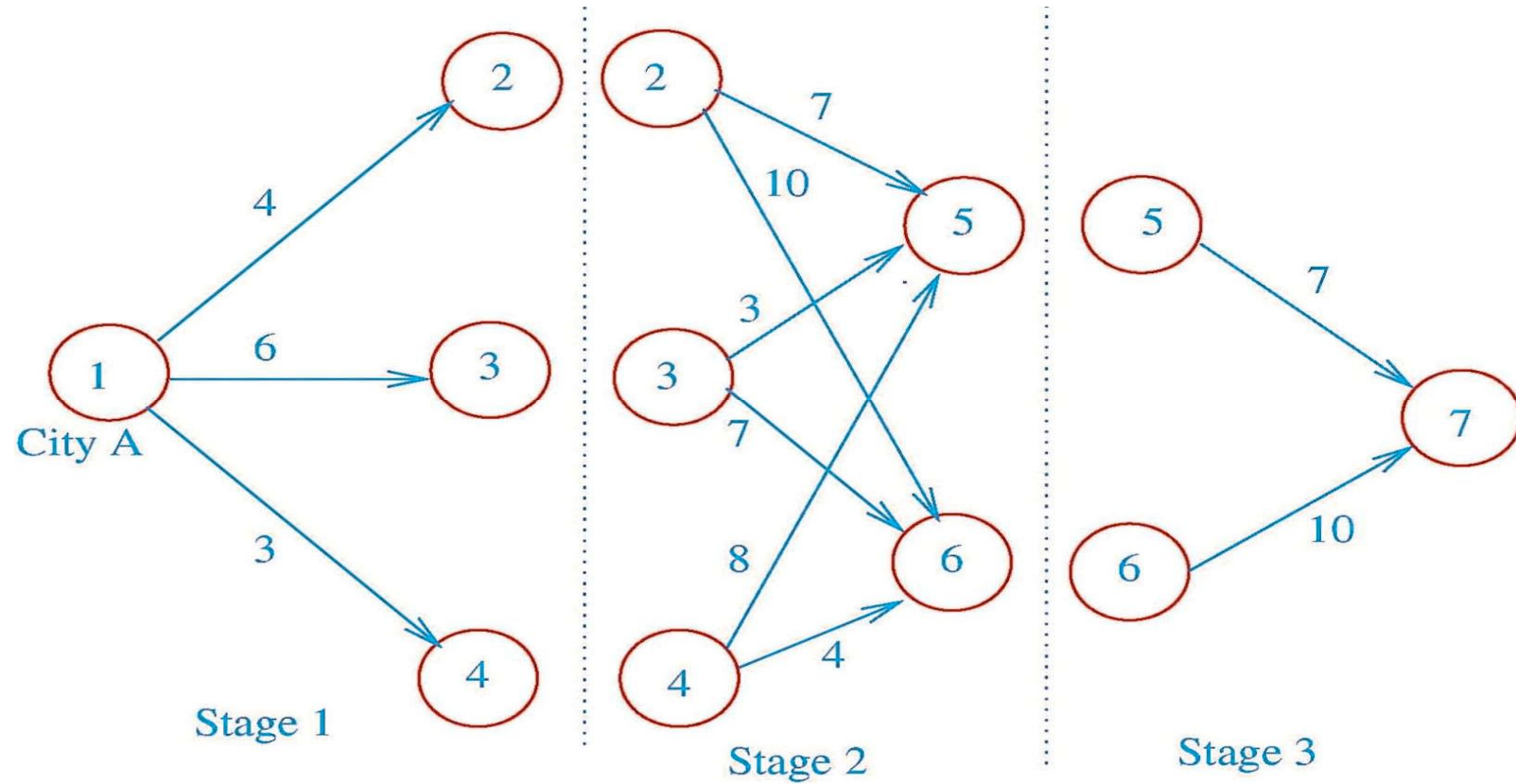


Example-1 : Solution

- ▶ To solve the problem by dynamic programming, we first decompose it into stages.
- ▶ We carry out the computations for each stage separately.



Solution...



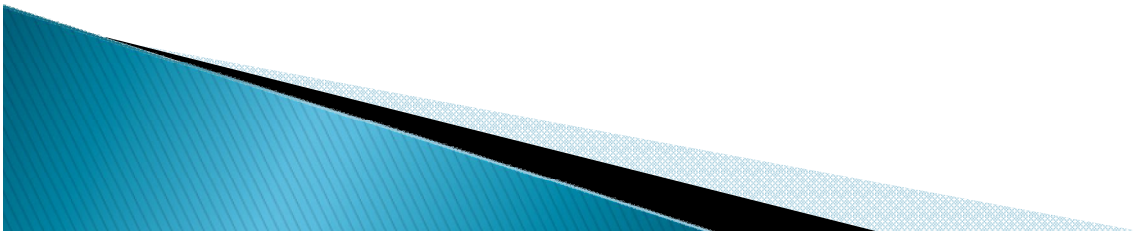
Stage 1:

We observe that stage 1 has three end nodes 2, 3 and 4. Therefore, from node 1

shortest distance to node 2 = 4 km,

shortest distance to node 3 = 6 km,

shortest distance to node 4 = 3 km.

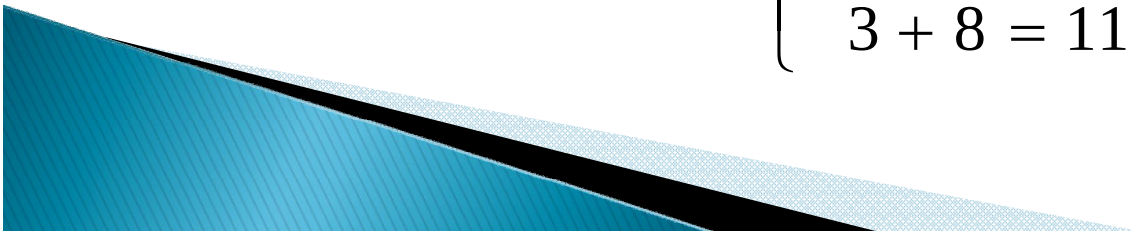


Stage 2:

In stage 2, we have two end nodes 5 and 6.

We calculate the shortest distance for each node as follows:

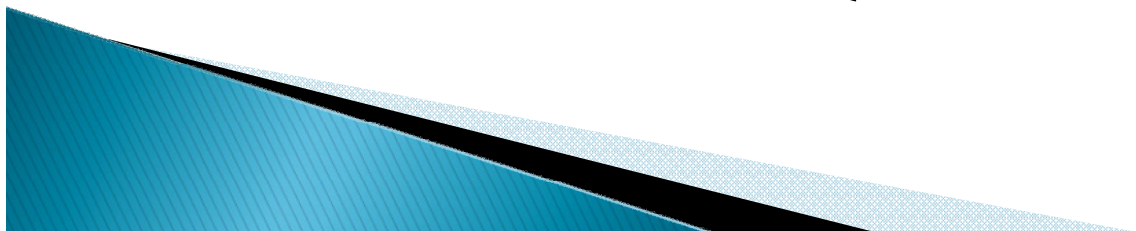
$$\begin{aligned} \left(\begin{array}{l} \text{shortest} \\ \text{distance} \\ \text{to node 5} \end{array} \right) &= \min_{i=2,3,4} \left\{ \left(\begin{array}{l} \text{shortest} \\ \text{distance} \\ \text{to node } i \end{array} \right) + \left(\begin{array}{l} \text{distance} \\ \text{from node} \\ \text{i to node 5} \end{array} \right) \right\} \\ &= \min \left\{ \begin{array}{l} 4 + 7 = 11 \\ 6 + 3 = 9 \\ 3 + 8 = 11 \end{array} \right\} = 9 \text{ (from node 3).} \end{aligned}$$



Stage 2...

Similarly, we calculate the shortest distance to node 6 as

$$\begin{aligned} \left(\begin{array}{l} \text{shortest} \\ \text{distance} \\ \text{to node 6} \end{array} \right) &= \min_{i=2,3,4} \left\{ \left(\begin{array}{l} \text{shortest} \\ \text{distance} \\ \text{to node } i \end{array} \right) + \left(\begin{array}{l} \text{distance} \\ \text{from node} \\ i \text{ to node 6} \end{array} \right) \right\} \\ &= \min \left\{ \begin{array}{l} 4 + 10 = 14 \\ 6 + 7 = 13 \\ 3 + 4 = 7 \end{array} \right\} = 7 \text{ (from node 4).} \end{aligned}$$

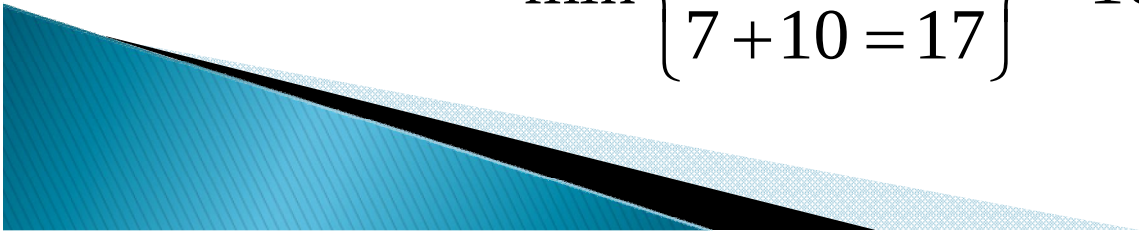


Stage 3:

In the last stage the destination node 7 can be reached from either node 5 or 6.

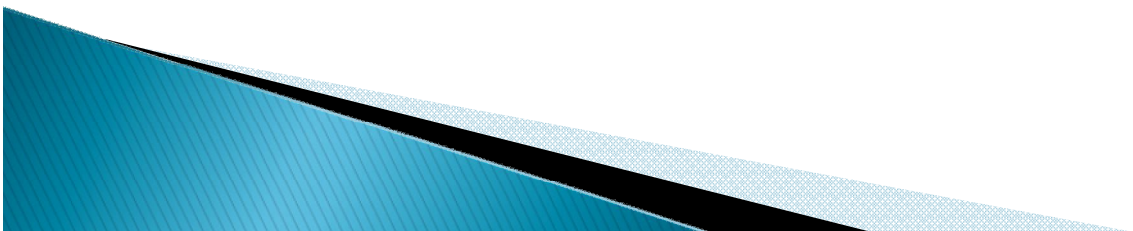
Using stage 1, and 2 results, we estimate the shortest distance to node 7 as follows:

$$\begin{pmatrix} \text{shortest} \\ \text{distance} \\ \text{to node 7} \end{pmatrix} = \min_{i=5,6} \left\{ \begin{pmatrix} \text{shortest} \\ \text{distance} \\ \text{to node } i \end{pmatrix} + \begin{pmatrix} \text{distance} \\ \text{from node} \\ i \text{ to node 7} \end{pmatrix} \right\}$$

$$= \min \left\{ \begin{array}{l} 9 + 7 = 16 \\ 7 + 10 = 17 \end{array} \right\} = 16 (\text{from node 5}).$$


Thus, the shortest distance between city A and city B is 16km.

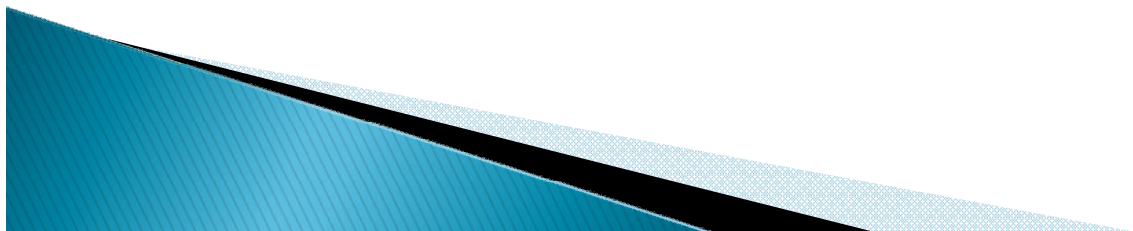
The shortest route is $1 \rightarrow 3 \rightarrow 5 \rightarrow 7$.

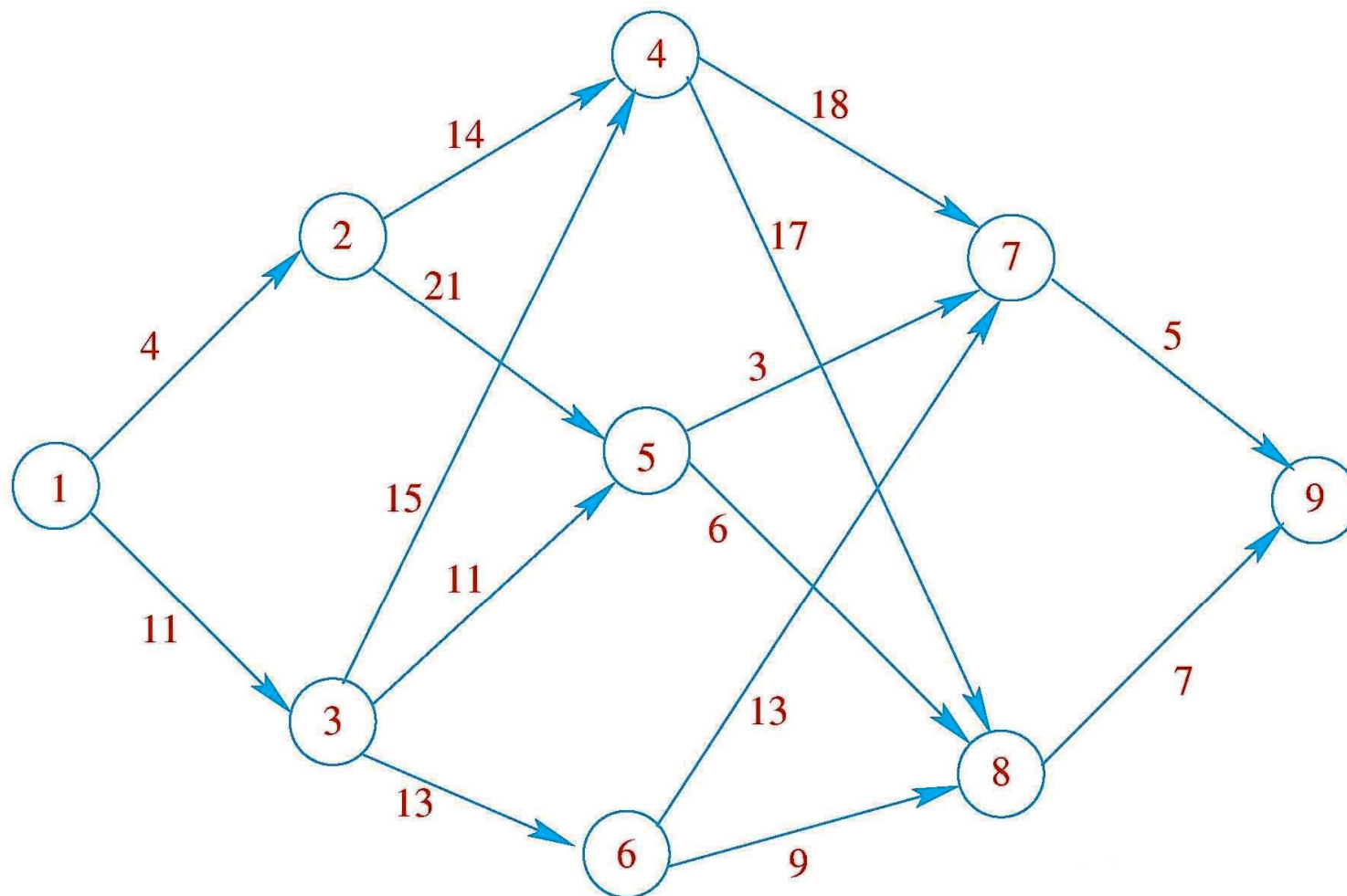


Example-2 (Shortest-Route Problem):

Find the shortest route between two cities.

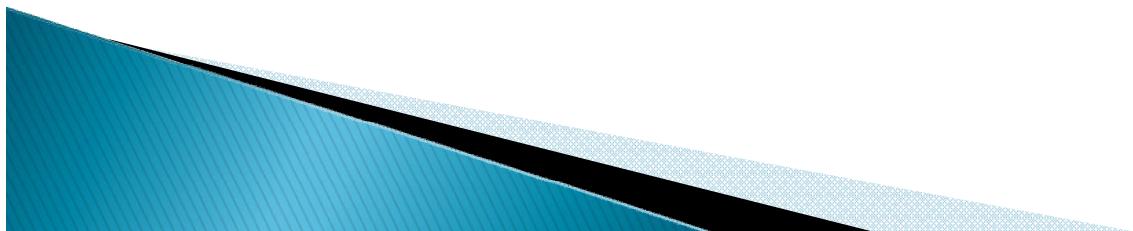
The below network provides the possible routes the starting city at node 1 and the destination city at node 9.

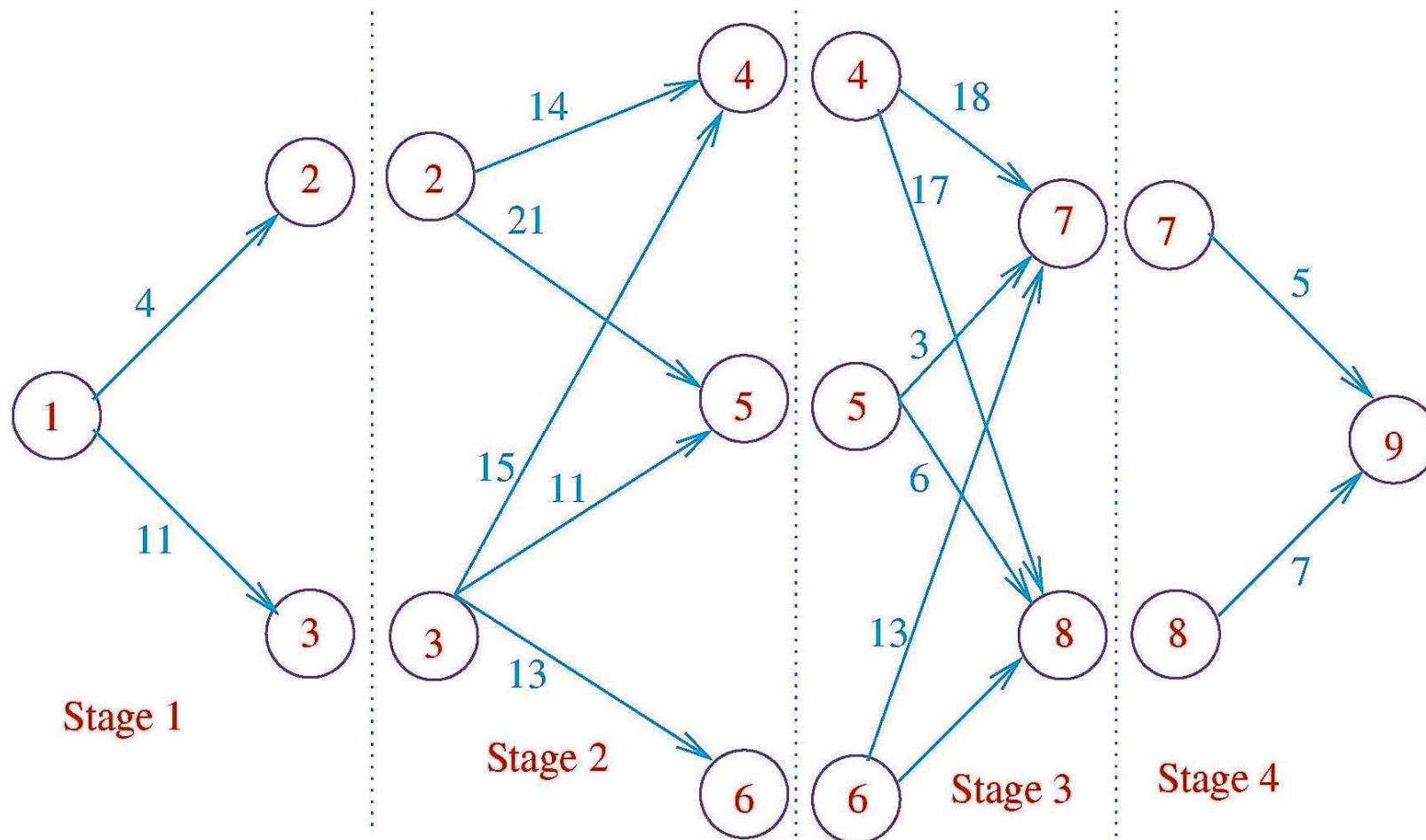




To solve the problem by dynamic programming (DP), we first decompose it into stages as delineated by the vertical dashed lines.

We carry out the computations for each stage separately.



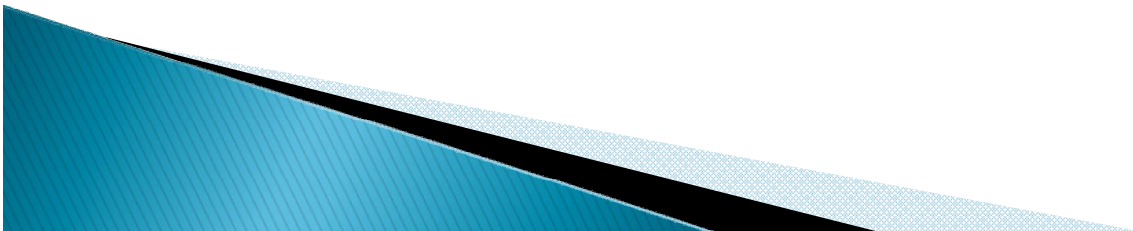


Stage-1

We observe that stage 1 has two end nodes 2 and 3. Therefore, from node 1

shortest distance to node 2 = 4 km,

shortest distance to node 3 = 11 km.

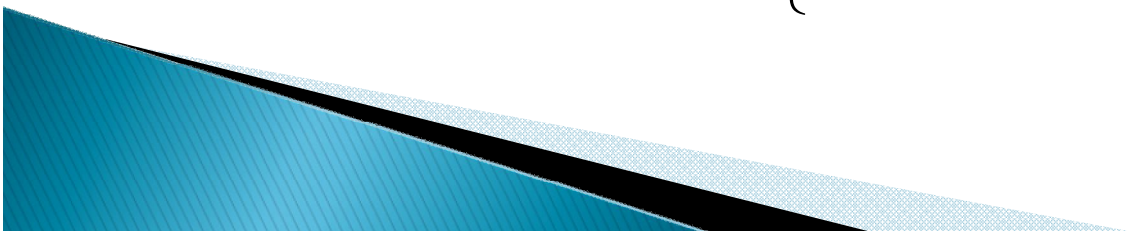


Stage-2

In stage 2, we have three end nodes 4, 5, and 6.

We calculate the shortest distance for each end node as follows:

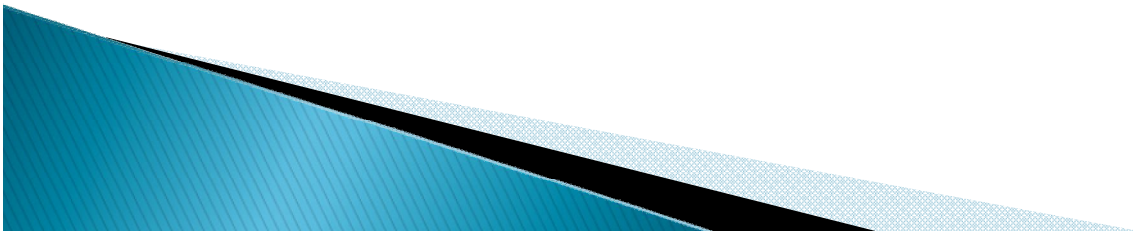
$$\begin{aligned} \left(\begin{array}{l} \text{shortest} \\ \text{distance} \\ \text{to node 4} \end{array} \right) &= \min_{i=2,3} \left\{ \left(\begin{array}{l} \text{shortest} \\ \text{distance} \\ \text{to node } i \end{array} \right) + \left(\begin{array}{l} \text{distance} \\ \text{from node} \\ i \text{ to node 4} \end{array} \right) \right\} \\ &= \min \left\{ \begin{array}{l} 4 + 14 = 18 \\ 11 + 15 = 26 \end{array} \right\} = 18 \text{ (from node 2).} \end{aligned}$$



Stage-2

- ▶ For node 5, we find that

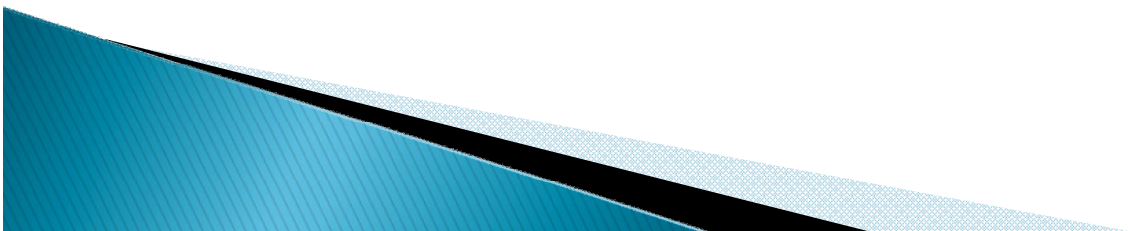
$$\begin{aligned} \left(\begin{array}{l} \text{shortest} \\ \text{distace} \\ \text{to node 5} \end{array} \right) &= \min_{i=2,3} \left\{ \left(\begin{array}{l} \text{shortest} \\ \text{distace} \\ \text{to node i} \end{array} \right) + \left(\begin{array}{l} \text{distace} \\ \text{from node} \\ \text{i to node 5} \end{array} \right) \right\} \\ &= \min \left\{ \begin{array}{l} 4 + 21 = 25 \\ 11 + 11 = 22 \end{array} \right\} = 22 \text{ (from node 3).} \end{aligned}$$



Stage-2

- ▶ Similarly, we calculate for node 6 as

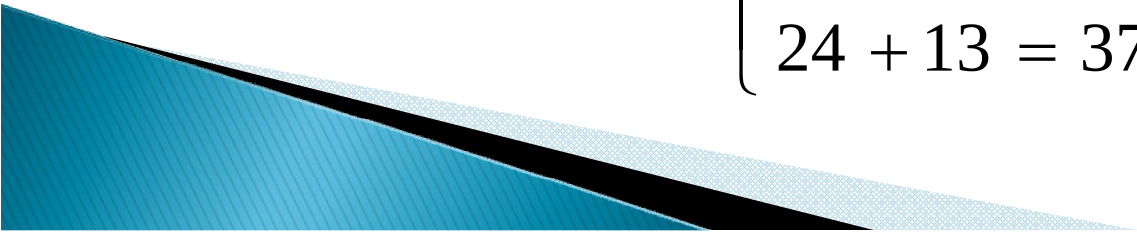
$$\begin{aligned} \left(\begin{array}{l} \text{shortest} \\ \text{distace} \\ \text{to node 6} \end{array} \right) &= \min_{i=3} \left\{ \left(\begin{array}{l} \text{shortest} \\ \text{distace} \\ \text{to node i} \end{array} \right) + \left(\begin{array}{l} \text{distace} \\ \text{from node} \\ \text{i to node 5} \end{array} \right) \right\} \\ &= 11 + 13 = 24 \text{ (from node 3).} \end{aligned}$$



Stage-3

In stage 3, we have two end nodes 7 and 8.

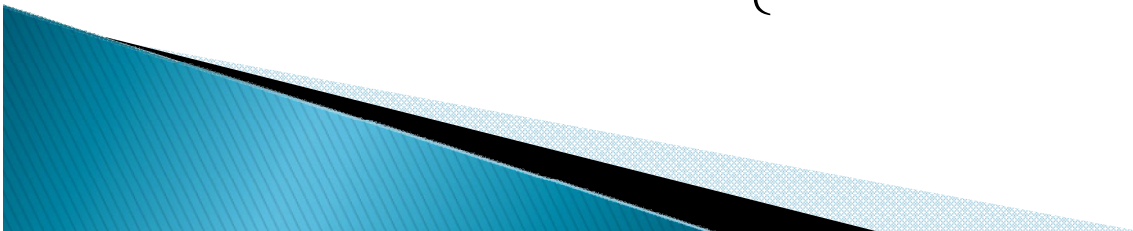
We calculate the shortest distance for each end node as follows:

$$\begin{aligned} \left(\begin{array}{l} \text{shortest} \\ \text{distance} \\ \text{to node 7} \end{array} \right) &= \min_{i=4,5,6} \left\{ \left(\begin{array}{l} \text{shortest} \\ \text{distance} \\ \text{to node i} \end{array} \right) + \left(\begin{array}{l} \text{distance} \\ \text{from node} \\ \text{i to node 7} \end{array} \right) \right\} \\ &= \min \left\{ \begin{array}{l} 18 + 18 = 36 \\ 22 + 3 = 25 \\ 24 + 13 = 37 \end{array} \right\} = 25 \text{ (from node 5).} \end{aligned}$$


Stage-3

- ▶ Similarly, the shortest distance to node 8 is

$$\begin{aligned} \left(\begin{array}{l} \text{shortest} \\ \text{distace} \\ \text{to node 8} \end{array} \right) &= \min_{i=4,5,6} \left\{ \left(\begin{array}{l} \text{shortest} \\ \text{distace} \\ \text{to node i} \end{array} \right) + \left(\begin{array}{l} \text{distace} \\ \text{from node} \\ \text{i to node 8} \end{array} \right) \right\} \\ &= \min \left\{ \begin{array}{l} 18 + 17 = 35 \\ 22 + 6 = 28 \\ 24 + 9 = 33 \end{array} \right\} = 28 \text{ (from node 5).} \end{aligned}$$



Stage-4

In the last stage the destination node 9 can be reached from either node 7 or 8.

Using stage 1, 2, 3 results, we estimate the shortest distance to 9 as follows:

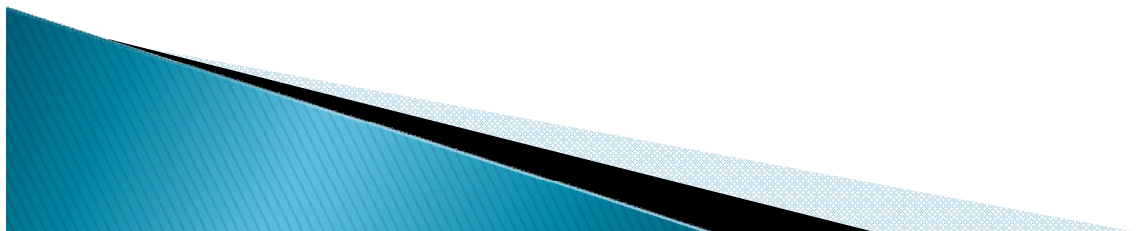
$$\left(\begin{array}{l} \text{shortest} \\ \text{distance} \\ \text{to node 9} \end{array} \right) = \min \left\{ \begin{array}{l} 25 + 5 = 30 \\ 28 + 7 = 35 \end{array} \right\} = 30 (\text{from node 7}).$$



Stage-4

Thus, the shortest distance between nodes 1 and 9 is 30km.

The shortest route is $1 \rightarrow 3 \rightarrow 5 \rightarrow 7 \rightarrow 9$:



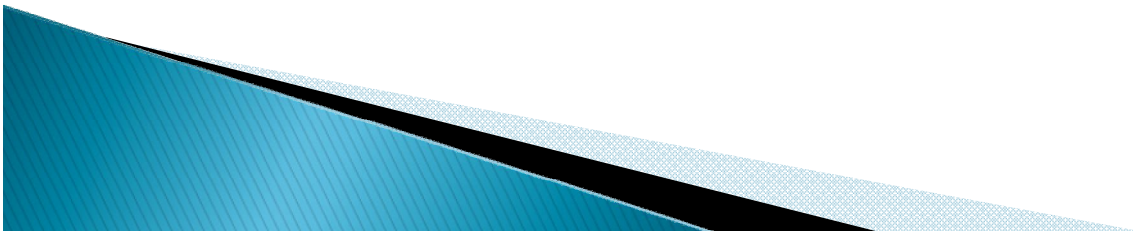
Formulation:

Let $f_i(x_i)$ be the shortest distance to node x_i at stage i and let $d(x_{i-1}, x_i)$ be the distance from node x_{i-1} to node x_i .

Then f_i is computed from f_{i-1} using the following recursive equation

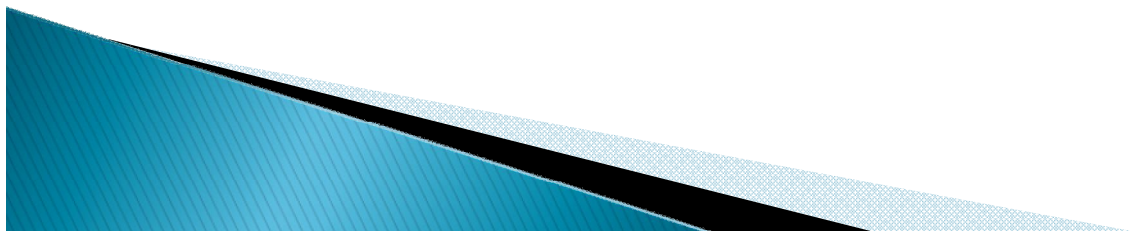
$$f_i(x_i) = \min_{\text{all feasible } (x_{i-1}, x_i) \text{ routes}} \{d(x_{i-1}, x_i) + f_{i-1}(x_{i-1})\}, i = 1, 2, 3$$

Starting at $i = 1$, the recursion sets $f_0(x_0) = 0$. This shows that the shortest distances $f_i(x_i)$ at stage i must be expressed in terms of the next node x_i .



Forward and Backward Recursion

- ▶ In *Forward recursion*, the computations proceed from left to right (from stage 1 to stage 3).
- ▶ In *Backward recursion*, the computations proceed from right to left (from stage 3 to stage 1).
- ▶ Both the forward and backward recursions yield the *same solution*.

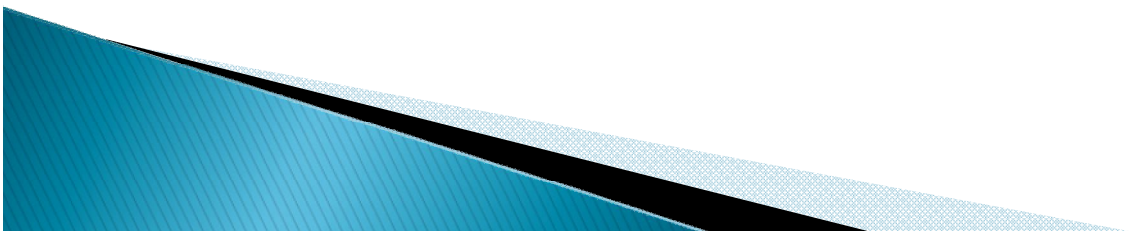


Backward Recursion

The backward recursive equation is

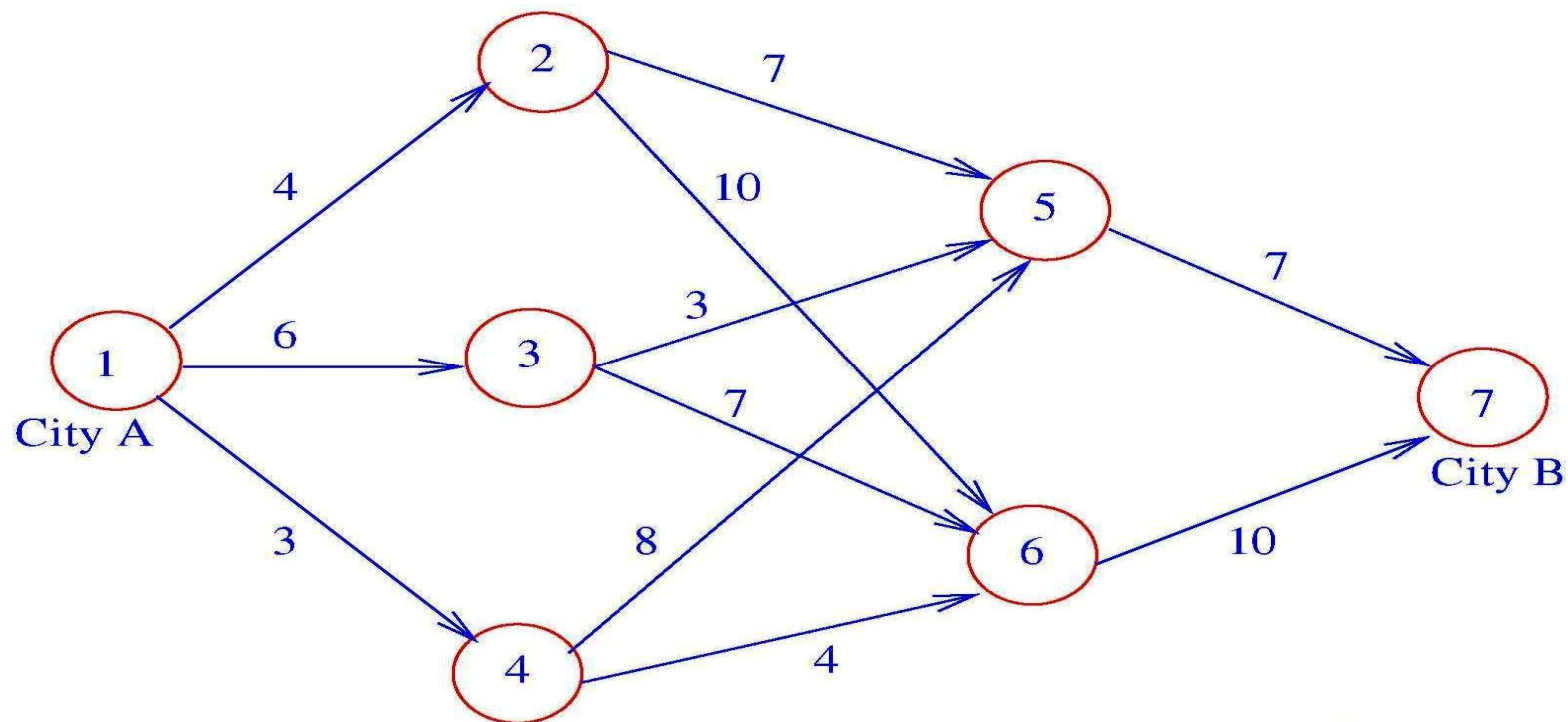
$$f_i(x_i) = \min_{\text{all feasible } (x_i, x_{i+1}) \text{ routes}} \{d(x_i, x_{i+1}) + f_{i+1}(x_{i+1})\}, i = 1, 2, 3$$

with $f_4(x_4) = 0$



Example :

- ▶ For the given network, develop the backward recursive equation and use it to find the optimum solution. The network as follows:



Solution

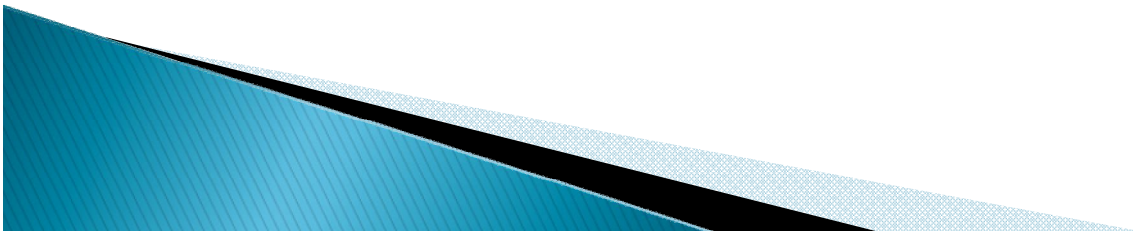
The backward recursive equation is

$$f_i(x_i) = \min_{\text{all feasible } (x_i, x_{i+1}) \text{ routes}} \{d(x_i, x_{i+1}) + f_{i+1}(x_{i+1})\}, i = 1, 2, 3$$

where, $f_4(x_4)=0$ for $x_4 = 7$.

The sequence of the computations is

$$f_3 \rightarrow f_2 \rightarrow f_1$$



Stage-3

We observe that node 7 ($x_4 = 7$) is connected to node 5 and 6 ($x_3 = 5, 6$) with exactly one route each.

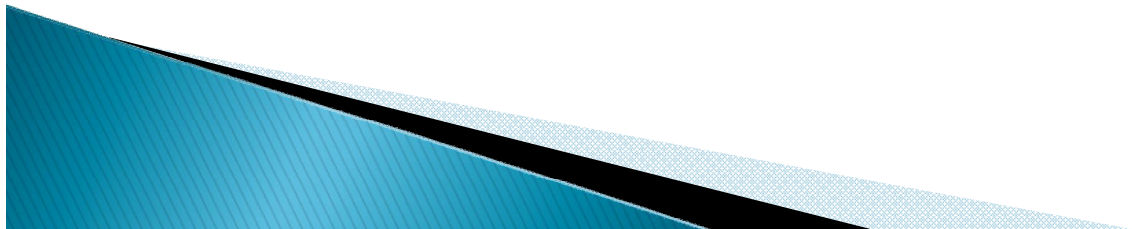
The computations are as follows:

x_3	$d(x_3, x_4)$	Optimum solution	
	$x_4 = 7$	$f_3(x_3)$	x_4^*
5	7	7	7
6	10	10	7

Stage 2

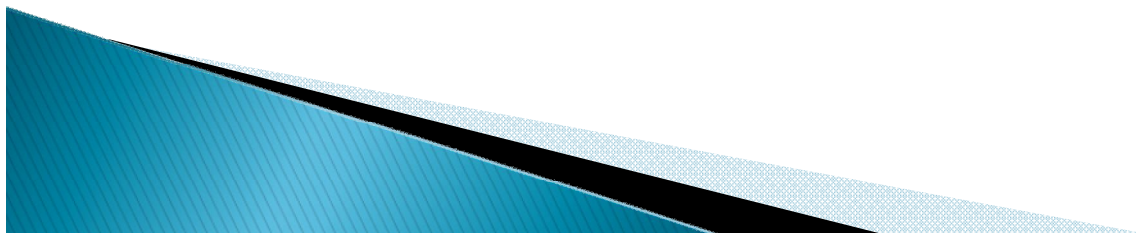
We observe that node 5 and 6 are connected to nodes 2, 3 and 4.
Using $f_3(x_3)$ from stage 3, we perform the computations as follows:

x_2	$d(x_2, x_3) + f_3(x_3)$		optimum solution	
	$x_3 = 5$	$x_3 = 6$	$f_2(x_2)$	x_3^*
2	$7+7=14$	$10+10=20$	14	5
3	$3+7=10$	$7+10=17$	10	5
4	$8+7=15$	$4+10=14$	14	6



The optimum solution of stage 2 as follows:

If you are in cities 2 or 3, the shortest route passes through city 5, and if you are in city 4, the shortest route passes through city 6.

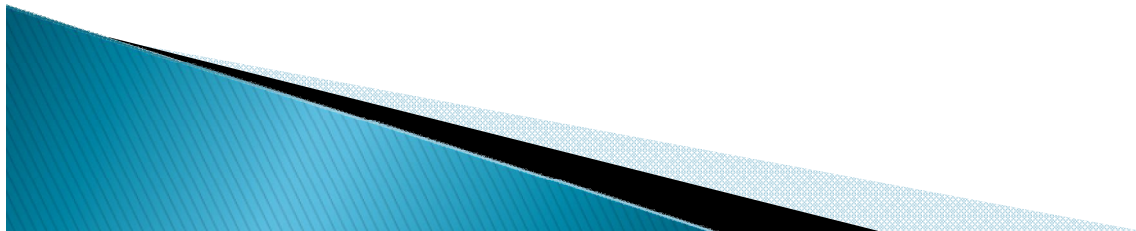


Stage-1

We note that node 1 is associated to nodes 2, 3 and 4.

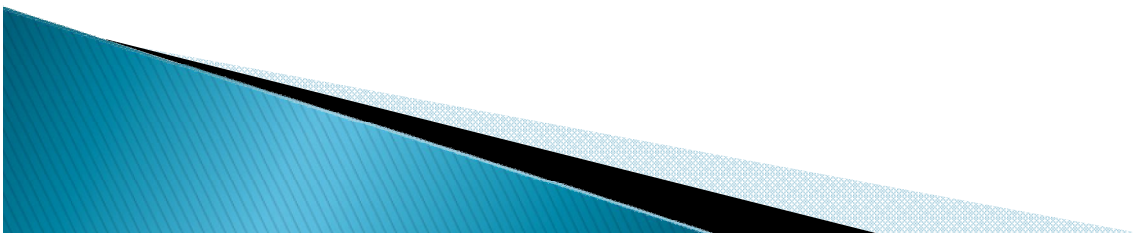
With the help of $f_2(x_2)$, we compute the shortest distance as

x_1	$d(x_1, x_2) + f_2(x_2)$			optimum solution	
	$x_2 = 2$	$x_2 = 3$	$x_2 = 4$	$f_1(x_1)$	x_2^*
1	4+14=18	6+10=16	3+14=17	16	3



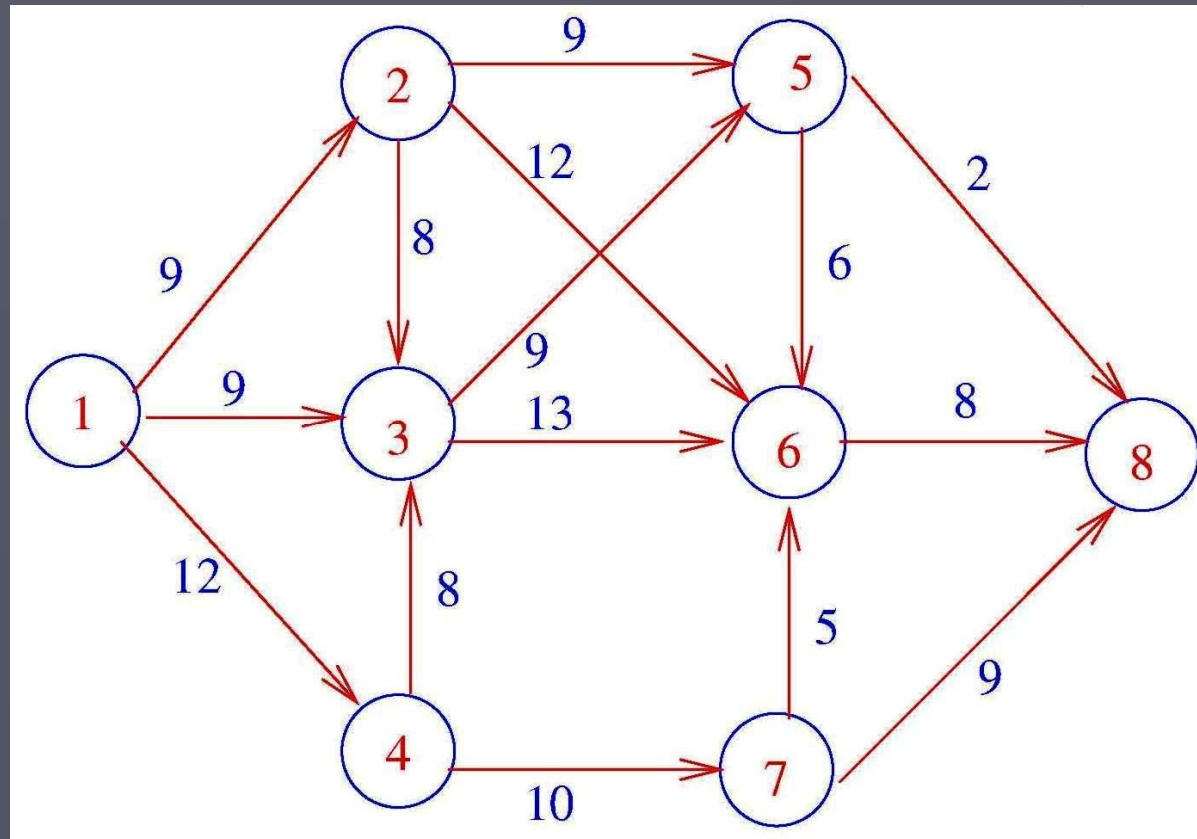
Therefore, the optimum solution at stage 1 is 16, i.e., the shortest distance between nodes 1 and 7 is 16 km.

- ▶ The shortest path is $1 \rightarrow 3 \rightarrow 5 \rightarrow 7$.



Problem 1:

Determine the shortest route between cities 1 and 8. Define the stages and the states using backward recursion and then solve the problem.



Problem: Use dynamic programming to

Minimize $z = y_1^2 + y_2^2 + y_3^2$

subject to $y_1 + y_2 + y_3 \geq 30,$

$$y_1, y_2, y_3 \geq 0$$

Solution: There are three stages: in stage i , we select the variable y_i . At stage i , we are in state x_i = the sum of the variables from stage i to n .

Thus

$$x_3 = y_3$$

$$x_2 = y_2 + y_3 \Rightarrow x_2 - y_2 = x_3$$

$$x_1 = y_1 + y_2 + y_3 \Rightarrow x_1 - y_1 = x_2$$

Let $f_i(x_i)$ = the optimal return for stages $i, i+1, \dots, 3$

$n = 3$: Here y_3 can take only one value, namely x_3
and so optimal return $f_3(x_3) = x_3^2$

$n = 2$: Here $f_2(x_2) = \min_{0 \leq y_2 \leq x_2} \{y_2^2 + f_3(x_2 - y_2)\}$
 $= \min_{0 \leq y_2 \leq x_2} \{y_2^2 + (x_2 - y_2)^2\}$

Using calculus, we find Optimal $y_2^* = \frac{x_2}{2}$

and $f_2(x_2) = \frac{x_2^2}{2}$

n = 1: Here

$$f_1(x_1) = \min_{0 \leq y_1 \leq x_1} \{y_1^2 + f_2(x_1 - y_1)\}$$
$$= \min_{0 \leq y_1 \leq x_1} \left\{ y_1^2 + \frac{(x_1 - y_1)^2}{2} \right\}$$

Using calculus, we find optimal

$$y_1^* = \frac{x_1}{3}$$

$$f_1(x_1) = \frac{x_1^2}{3}$$

Since $x_1 \geq 30$, $f_1(x_1)$ is minimum when $x_1 = 30$

Thus min value of the problem = 300 and is got when

$$y_1^* = 10, y_2^* = 10, y_3^* = 10.$$